

KHANPUR, Pakistan

WMO: 417180

Lat: 28.650N

Lon: 70.683E

Elev: 88

StdP: 100.27

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.2	5.4	0.4	3.9	13.2	1.9	4.4	12.6	5.2	18.4	4.3	17.9	0.0	N/A	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.9	44.0	25.4	42.5	25.5	41.1	25.9	30.1	37.6	29.7	37.1	29.3	36.6	1.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.5	25.1	34.6	28.0	24.4	34.4	27.5	23.7	34.1	102.1	38.0	99.6	37.0	97.6	37.0	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.5	5.1	4.1	DB	2.0	46.5	1.5	1.2	1.0	47.3	0.1	48.0	-0.7	48.7	-1.8	49.6	
			WB	1.8	31.3	1.7	0.4	0.6	31.6	-0.4	31.9	-1.4	32.1	-2.6	32.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.1	13.4	17.0	22.8	29.7	34.2	35.1	34.1	32.5	30.9	27.1	20.7	15.1	(6)
		DBStd	7.99	2.30	2.99	3.46	3.19	2.29	1.84	1.72	1.33	1.55	2.37	2.44	2.52	(7)
		HDD10.0	3	3	0	0	0	0	0	0	0	0	0	0	1	(8)
		HDD18.3	325	154	56	6	0	0	0	0	0	0	0	5	104	(9)
		CDD10.0	5875	107	196	397	590	751	752	748	697	626	531	322	158	(10)
		CDD18.3	3156	0	19	144	340	493	502	490	439	376	272	77	4	(11)
		CDH23.3	45489	19	224	1461	4549	7782	8138	7791	6516	5131	3038	758	83	(12)
(13)		CDH26.7	28907	0	51	666	2843	5388	5752	5322	4045	2931	1662	240	6	(13)
(14)	Wind	WSAvg	0.9	0.5	0.9	1.0	1.1	1.3	1.7	1.5	1.2	0.9	0.4	0.4	0.4	(14)
(15)	Precipitation	PrecAvg	88	5	3	5	3	4	29	27	11	0	1	4		(15)
(16)		PrecMax	263	28	9	30	10	22	40	112	148	92	2	5	27	(16)
(17)		PrecMin	15	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	68	7	3	9	3	6	9	37	39	25	0	1	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	25.4	30.6	37.9	44.1	46.0	45.2	42.2	39.4	39.4	37.7	32.7	27.6		(19)
(20)		0.4% MCWB	15.2	18.2	22.0	22.6	24.8	26.6	28.5	28.7	28.1	25.1	21.2	18.3		(20)
(21)		2% DB	23.3	28.2	35.0	41.9	44.5	43.5	40.7	38.5	38.1	36.7	31.2	25.5		(21)
(22)		2% MCWB	14.6	17.9	21.3	21.7	25.1	26.8	28.9	28.6	27.7	24.8	21.0	16.7		(22)
(23)		5% DB	21.7	26.1	32.8	40.0	42.9	42.1	39.8	37.7	37.1	35.5	29.4	23.8		(23)
(24)		5% MCWB	14.0	17.3	20.5	21.7	25.0	27.2	29.0	28.5	27.4	24.3	20.1	15.6		(24)
(25)		10% DB	20.0	24.1	30.5	38.0	41.3	40.7	38.8	36.9	36.1	34.1	27.7	22.1		(25)
(26)		10% MCWB	13.1	16.2	19.8	21.8	24.9	27.5	28.9	28.5	26.9	23.5	19.0	14.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	16.5	20.7	24.2	25.8	29.4	30.6	30.8	30.2	29.9	27.5	23.1	19.1		(27)
(28)		0.4% MCDB	22.5	26.9	34.6	38.0	40.0	39.1	38.3	36.8	36.9	35.2	30.6	26.0		(28)
(29)		2% WB	15.6	19.4	22.7	24.4	28.2	29.7	30.3	29.7	29.2	26.5	21.8	17.7		(29)
(30)		2% MCDB	21.6	25.6	32.4	35.8	39.4	38.8	37.7	36.1	35.6	33.8	29.2	24.0		(30)
(31)		5% WB	14.7	18.2	21.6	23.6	27.4	29.3	29.8	29.3	28.6	25.6	21.0	16.6		(31)
(32)		5% MCDB	20.6	24.7	30.2	35.4	38.6	38.4	37.1	35.6	34.7	32.8	28.3	22.5		(32)
(33)		10% WB	13.9	16.9	20.5	22.9	26.6	28.7	29.4	28.9	27.9	24.8	20.0	15.3		(33)
(34)		10% MCDB	19.0	22.8	28.5	34.3	37.6	37.9	36.7	35.1	34.1	32.1	26.4	20.8		(34)
(35)	Mean Daily Temperature Range	MDBR	13.3	13.9	13.9	15.4	14.5	11.9	9.5	8.9	10.5	14.1	14.5	14.2		(35)
(36)		5% DB MCDBR	15.9	16.2	16.7	17.9	16.4	13.9	10.5	10.1	11.9	14.7	15.9	16.3		(36)
(37)		5% WB MCWBR	8.2	7.1	5.7	4.9	5.2	4.6	3.6	3.5	4.0	5.2	7.1	8.5		(37)
(38)		5% WB MCDBR	14.1	14.1	15.3	15.9	14.3	11.5	9.9	9.4	10.3	13.1	14.5	14.9		(38)
(39)		5% WB MCWBR	7.7	6.8	5.8	5.6	5.5	4.6	3.9	3.7	4.0	5.2	7.0	8.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.484	0.490	0.525	0.599	0.663	0.756	0.811	0.719	0.603	0.577	0.573	0.527		(40)
(41)		taud	1.859	1.887	1.824	1.700	1.608	1.501	1.450	1.595	1.767	1.723	1.693	1.774		(41)
(42)		Ebn at Noon	722	766	770	728	684	620	584	635	697	676	630	653		(42)
(43)		Edh at Noon	174	183	207	242	266	295	309	265	217	213	204	182		(43)
(44)	All-Sky Solar Radiation	RadAvg	3.67	4.62	5.88	6.74	7.17	6.89	6.38	6.15	5.93	5.25	4.08	3.54		(44)
(45)		RadStd	0.22	0.30	0.26	0.19	0.14	0.19	0.25	0.28	0.24	0.21	0.21	0.14		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page